SECTION A - Data Fundamentals.

1). What is a database?

- its a place where information is stored and it can be easily accessed and used.

- Database is essential in modern world because:
- it can store large amount of information
  e.g - Database in government depti can assist with storing millions of records, far better than manual records'
- Easily accessible. / easy fast access.

e.g Organisations are able to access data in seconds instead of hours.

2). What is Data warehouse?.

- its a type of database that stores large amounts of historical data from different sources in one place.
- For example data warehouse-is like a big archieve library where all past records are stored for analysis.
- Difference between Database and Data warehouse.
- Database - is for handling daily activities - placing orders daily
- Data warehouse - helps answer big questions like e.g order trends over months and years.

3). What is Data Lake?.

- large storage system that keeps all types of data in their raw form until you need to use it. eg excel files

- Explain how it differs from a data warehouse.
- Data lake stores everything now and figure it out later, whereas
- Data warehouse clean and organise first, then use it later

- A business prefers data lake when:
- they have lots of different types of data.
- they need low-cost storage for big data.

4) Data Mart
- is smaller, focused part of data warehouse designed for a specific team or department.
- for example - If data warehouse is a big library, then data mart is one section of that library. e.g Finance Dept.
- How does it relate to Data Warehouse?
- A data mart usually come from a data warehouse and its
- used by specific team like e.g supply chain, finance.

5) Structured Data.
- its kind/type of data that is easy to search and analyse, its organised in rows and columns.
- For example - A table will have a name, age, email organised rows and column.
- Semi-structured Data.
- type of data that is not fully organised but has more structure. its format its usually flexible e.g xml.
- Unstructured Data.
- type of data that has no clear structure, hard to organise and analyse e.g Instagram posts.

6) Data Modelling.
- its the process of planning how data will be organised, stored and related in the system.
- Example - its like a blueprint for data before building a system.

- Why Data Modelling important?
- it keeps data organised and easy to use.
- It avoids errors and duplication.
- Makes system faster and efficient.

- Types of Data Model.
- Conceptual - big picture view - Customers orders.
- Logicall - more detailed view - Customer's ID, name, Contacts.
- Physical - Very detailed - e.g it shows data in Database.

8) What is a Database management System?

- A Database management system (DBMS) is software that helps you store, manage, and retriere data easily and safely.

Popular Data Management system Platforms are:

- MySQL - Simple, Fast, widely used.

- PostgreSQL - Its powerful and supports complex queries
  e.g - its used for financial system

- Oracle Database - its very secure and scalable
  e.g - its used by large companies (banks).

- Mongo DB - its flexible, it store data like JSON
  use - Apps with changing data like Instagram.

9) What is ETL (Extract, Transform, Load)?

- ETL is a process used to collect data, clean it, and move it into a system for analysis.
- The three steps.

1. Extract.

- Get data from different sources (Database, files)
- For example - pull financial data from a website and excel file

2. Transform

- Clean and prepare the data by fixing errors, removing duplicates
  For example - fix missing value, convert dates.

3. Load.

- Store clean data in a system like a database.
  For example - load cleaned data into database.

- Why is ETL (Extract, Transform, Load) important?. Because
- it ensures that data is clean and reliable
- Combines data from multiple sources
- makes data ready for analysis and reporting.

10. What are common data storage formats used in analystics?

1) CSV (comma-separated values)

- this one is a very simple and is widely supported
- It consists of plain text, rows and columns.

Use

- its used when sharing data between tools (Excel)

2) JSON - (JavaScript Object Notation)

- this one is flexible and can handle nested data.

Use

- It is used when working with APIs (Applicat programming Interfa

3) Parquet -

- It is very highly compressed data storage Format, it is column-based and very fast for analytics.

Use

- it is used when working with big data.
- When running queries in tools like spark.

4. Avro.

- It is good for data streaming, its row-based, compact and supports schemas

Use

- when moving data between systems.
- When streaming.

SECTION B : Artificial Intelligence and machine Learning.

11) What is Artificial Intelligence (AI)

- It is when machines or software are designed to think, learn, and make decisions like humans.

- Types of AI

1. Narrow AI (weak AI)

- Its designed to do one specific task, cannot go beyond what is programmed for.

E.g - Goggle maps - Its designed only to find fastest routes and predict traffic cannot go beyond that.

2. General AI (Strong AI)

- Can think, learn and perform any task like a human
- It is not yet fully developed.

Example - A human like robot performing tasks like a human.

12) What is machine Learning (ML)?

- Machine learning is a type of AI where computers learn patterns from data and improve automatically, without being programmed step by step.

- Machine learning VS Traditional Programming.

- In Traditional learning - you write rules, the the system will follow the rules to get output.
- Whereas Machine Learning - you give data, then the system learns the rules and patterns itself to get the output.

- Types of Machine learning

1) Supervised learning - type of machine learning where a model learns from labeled data.

E.g you teach the system using examples with answers, so it can learn and predict new results.

2. Unsupervised Learning - is a type of machine learning where the model works with data that has no labels. (no correct answers).

Example - grouping customers on buying behaviour.

3. Reinforcement Learning - is a type of learning where an AI learns by trial and error and improves by recieving rewards or penalties.

E.g. - the system get rewarded everytime it get ~~correct~~ ~~AI~~ information correct and gets penalised when it predicts wrong information.

13.) What is deep learning?

- Is a type of machine learning that uses multi-layered neural networks to learn from large amounts of data.

e.g of data (Images, speech, text)

- How Deep learning relates to AI and ML

Lets look at it like a hierarchy.

- Artificial Intelligence = the big idea of machines acting smart
- Machine Learning = a way to achieve AI by learning from data
- Deep learning = a special type of ML that uses deep neural networks.
- What are Neural Networks?
- is a system designed to mimic how the Human brain works.

- How Neural Networks work:

1. Input data goes in - e.g image of a flower.
2. Hidden layers process it - e.g identify shapes, patterns
3. Output is then produced - e.g it will say flower or not flower.
4. Learning happens - the system checks if it was correct.
   - If wrong, it adjusts itself.

14.) What is Natural Language processing (NLP)?

- Natural language processing is a branch of AI that helps computers understand, interpret and respond to human language (like English, sepedi).

- NPL in short - making computers understand human language.

- Real world applications of NLP.

Example - Google Translate. - it translate text and speech between English and sepedi language

15). What is large language Model (LLM)?

- A large Language Model is an advanced AI Model trained on huge amounts of text data to understand and generate human-like language.
- For example LLM can
  - answer questions
  - write text and summarize information.

- Difference between LLM ~~vs~~ and Standard Machine Learing Model.

Standard Learning Model	Large Language Model (LLM)
• Built for a specific task	• Can do many language tasks in one model.
• Uses structured data.	• Built to understand human language.
•	• ChatGPT is an example of LLM

16). What is Generative AI ?

- Generative AI is a type of AI that can create new content such as text, images, music or code.
- For example - Instead of analysing data, it generates something new.

- How it works at high level - it works by :

- Being trained on large amounts of data.
- by learning patterns and using those patterns to generate new content.

- Three Generative AI Tools.

1. ChatGPT - used to generate text, answers and essays.
   e.g - It can be used for writing help.

2. DALL.E - Its used to generate images from text descriptions
   for e.g - a dog wearing a hat.

3. GitHub Copilot - its used to generate code suggestions.
   - It helps developers write code faster.

17.) What is a Training Dataset?

- A training dataset is the data used to teach an AI model how to make decisions or predictions.

- Why is data quality important?
- Good data produces accurate, reliable results, whereas
- Bad data produces wrong, misleading results, we therefore require good data quality to produces accurate outputs.

- What can go wrong with bad or biased data?

1. Wrong predictions - If data has errors it gives incorrect results
2. Bias and unfairness - If data favours one group, then the model becomes unfair.
   e.g - Using AI only to screen job application.

3. Poor Performance.
- Model may fail in real-world situations.

18) What is Computer Vision?

- Computer vision is a field of AI that enables machines to see, understand, and interpret images or videos.

- How machines 'see' images - by:

1.) Converting images into numbers and then find patterns in those numbers.
2.) By detecting features - so the system will detect patterns like lines, edges, colors etc.
3.) By recognising patterns - so the system learns what those patterns mean. e.g being able to recognise a face.
4.) By making predictions - the system is able to label or interprets the images

19.) What is an API (Application Programming Interface)?

- API (Application Programming Interface is a way for different software systems to communicate and share data.

Analogy Explanation.

- For example think of an API like a TV remote control
- You press the button (request), the remote (API) sends signal to TV (system)
- The TV responds (change channel, volume etc), so bottom line
- You dont need to know how TV works inside - you just use the remote.

- Why APIs are important - because
- they connect systems - apps can talk to each other.
- Save time - reuse existing services
- Enable integration - combine services (payments etc)
- Access data easily by getting real-time information

20) What is an API key?

- An API key is a unique secret code used to identify and authenticate a user or application when accessing an API.
- Its like a password for using an API

- Purpose of API key.
- its to identify who is making the request.
- its to control ~~ano~~ access to the API

- How it should be stored and protected?
- By keeping it private (never share publicly)
- Keep your API like a password
- Store in secure, private locations like enviroment variables

• What happens if API key is exposed?

• Because API key is a secret identifier for accessing APIs, so if exposed, it can lead to misuse, costs and security risks.

21.) What is a Rest API?

• A REST API is a type of API that allows system to communicate over the web using standard HTTP methods.

• What does REST stand for?
REST - ~~Standard~~ Representational State Transfer.

• Main HTTP Methods and its uses:

1) GET - used to retrieve data. e.g retrieving a list of suppliers
2) POST - is used to create new data. e.g Adding a new supplier
3) PUT - its used for updating data e.g updating supplier's details.
4) DELETE - its used to remove data e.g Deleting supplier.

22). What is JSON?

• JSON (JavaScript Object Notation) is a simple format used to store and exchange data between systems.

• Why is it commonly used? Because
. it's easy to read and write
. its lightweigh and works with most programming language

• Example of JSON
. "name" → Thabitha.
. "age" → 37

23) What is prompt Engineering?

. Prompt engineering is the skill of writing clear and effective instructions to get better results from AI tools such as CHATGPT.

. Why is it important? Because

. it creates better prompts that gives better answers.

. It also save time, to avoid less back and forth

. it helps AI give accurate and detailed responses.

. Examples of poorly written prompt and improved version.

1. "Tell me about South Africa" - poor prompt.
"Explain a short overview of South Africa including it population cities and economy" - Better prompt.

2. Poor prompt - "write something about Data Analytics"
Better prompt - "Explain what Data Analytics is and why it is important in business"

24) What is a map function?

. A map function is used to apply a specific operation to each item in a list and return a new list with the results.

. What MAP function does in programming and data processing.

. It takes a list, applies a rule and then returns a new transformed list.

. Example in Python.

. Original list - 5, 10, 15, 20

. Rule - multiply each number by 2.

. New list - 10, 20, 30, 40.

25). What is cloud computing?

- Cloud computing is the delivery of computing services (eg servers) over the internet.
- it allows you to access everything online, anytime.

- Three main Cloud service models.

1) IaaS (Infrastructure as a service)
- With this model you rent basic IT infrastructure over the internet, instead of buying and maintaining physical hardware.
- Amazon web Services (AWS) is an example of IaaS.

2) PaaS (Platform as a service)
- its a ready made platform to build, test and deploy applications
- the enduser is responsible for managing the App and data
- whereas provider manages, servers, infrastructure etc.
- An example of PaaS is - Google App Engine

3) SaaS (Software as a service)
- it is fully ready software you use over the internet.
- the provider manages everything, and the end user just use.
- An example of SaaS in this case - Microsoft 365 (emails etc)

26) What is version control system?

- A version control system is a tool that helps you track changes in files over time.
- In simple, it allows you to save different versions of your work,
- go back to previous version if needed, and
- work safely in teams without overwriting work.

- Git - is a popular version control system used to track and manage changes in code and data projects.

- Git is used because :
- its fast and reliable
- its good for teamwork and
- it keeps a full history of changes.

- Why developers and data analysts use Git? they use to:
- Track changes in code and analysis script
- to avoid losing their work
- to collaborate with other safely.

- What is "commit" and "Branch"
- A commit is a saved snapshot of your work at a specific time

- A Branch is a separate version of your project used for working on new features or experiments.
- Example : Building / upgrading a SAP 4HANA system.
Main branch will be - live / production system
New branch will be - testing / dummy system.

27.) What is a Jupyter Notebook?
- Jupyter Notebook is an interactive tool used to write and run code step by step.

- What programming language is commonly used?
- Phyton is commonly used.

- Why is it popular for Data Analysts and Data scientists? Beca
- its good for testing ideas quickly
- it easily creates charts and graphs.
- its good / helps in data presentations.

28.) What is Python?

. Python is a high level programming language used to write instructions for computers in a simple and readable way.

• It is mainly used for analysing data, web development. etc.

• Why Python is popular for Data Analytics & AI.

. Because :

- its easy to learn and read.
- Works well with big data and machine learning

• Common Python libraries in Data work.

1. NumPy - it's used for math and working with numbers in bulk
2. Pandas - it's used for organising and analysing data in tables.
3. Matplotlib - it's used to create graphs and charts.
4. Scikit-learn - it's used for building simple machine learning models.

29) What is Data Privacy?

. Data privacy is about protecting personal Information like (email, ID, location etc).

• Why is Data privacy important?

. To protect people from identity theft and fraud.
. To ensure companies handle data legally and ethically
. To prevent unauthorised access to personal data.
. To build trust between users and companies.

• What is GDPR?

. GDPR (General Data Protection Regulation),
. is a data privacy law created by the European Union to protect personal data and privacy.

- The idea behind GDPR is to ensure clear consent is given by user before data is collected and for people to be able to access and update data.

- Which countries does GDPR apply to?
- All European countries.
- And some companies outside of EU, especially those doing business with EU countries.

30). What is Data Ethics.
- Data ethics refers to the rules and principles that guide how data and AI should be collected, used and shared in a fair, responsible and transparent way.

- Examples of unethical use of data/AI

1. Selling personal data without consent.
- Selling personal data to advertisers without informing users.
- The problem is that, it breaks trust, privacy and transparency.

2. Biased AI Hiring System.
- When a company uses AI to screen job applications, but the system is trained on past hiring data that mostly includes men, then
- the problem is, AI might unfairly reject qualified woman.

31.) What is Data Governance?
- Data governance is rules, processes on how a company controls and manages its data.

• Key components of a good data governance framework.

1. Data policies and standards.
• governs on how data should be collected, stored and used.

2. Compliance and regulations
• Ensure business follows laws, to avoid legal issues.

3. Data lifecycle management.
• manages data from creation to deletion

4. Data quality
• Ensures data is accurate and reliable

5. Data security and privacy.
• Protects data from unauthorised access.

32). What is Bias in AI
• Bias in AI happens when system produces unfair results because of how it was trained.

• What it means for a model to be biased.
• It mean AI learns certain pattern on how its trained for example if its trained on only recognising women's voice, it will learn that pattern, and when it recognises a different voice like a mans it produces unfair results.

• Why is this a problem.
• It becomes a problem since it leads to discrimination. and it creates harm for people's opportunities like eig job ~~dis~~ application disqualification.

Key Differences Between Data roles

Role	Focus	Responsibilities	Key Skills	Tools.
1. Data Analyst	is to interpret data.	DA are responsible for cleaning and analysing data.	SQL Data visualisation Excel	Power BI Python SQL
2. Data Engineer	Building systems to move and store data	manage databases and warehouses	Cloud SQL	AWS Apache spark Azure.
3. Data Scientist	DS focus mainly on predicting what will happen.	Responsible for performing statistical analysis.	Statistics Python	~~Pyy~~ Python Jupyter Notebooks.
4. AI Engineer	Deploying AI systems into production.	Integrate models into applications.	Software engineering APIs	Fast API PyTorch Docker.

34). What is Business Intelligence (BI)

- Business Intelligence is the process of collecting, analysing and visualising data to help businesses make better decisions.
- It turns raw data into reports, dashboards and trends.

- Popular BI Tools.

1. Power BI - It creates interactive dashboards and reports.
   - It helps track sales, performance, and KPI in real time.
2. Tableau - its a tool for data visualisation and storytelling
   - it turns complex data into easy to understand charts
   - it helps businesses understand trends.